

Introduction to Bayesian Statistics

SMIP Summer School 2025: Lecture 03

The Basics of Bayesian Analyses

- Bayesian statistical analysis refers to the use of models where some or all of the parameters are treated as random components
 - Each parameter comes from some type of distribution
- The likelihood function of the data is then augmented with an additional term that represents the likelihood of the prior distribution for each parameter
 - Think of this as saying each parameter has a certain likelihood – the height of the prior distribution
- The final estimates are then considered summaries of the posterior distribution of the parameter, conditional on the data
 - In practice, we use these estimates to make inferences, just as is done when using non-Bayesian approaches (e.g., maximum likelihood/least squares)

Why are Bayesian Methods Used?

- Bayesian methods get used because of the *relative* accessibility of one method of estimation (MCMC – to be discussed shortly)
- There are four main reasons why people use MCMC:
 1. Missing data
 2. Lack of software capable of handling large sized analyses (e.g., computational speed)
 3. New models/generalizations of models not available in software
 4. Philosophical reasons (e.g., Bayesian ideals)

Perceptions and Issues with Bayesian Methods

- The use of Bayesian statistics has been controversial
 - The use of certain prior distributions can produce results that are biased or reflect subjective judgment rather than objective science
- Most MCMC estimation methods are computationally intensive
 - Many “easy” Bayesian programs take a lot of syntax
- Understanding of what Bayesian methods had been very limited outside the field of mathematical statistics
- Over the past 20 years, Bayesian methods have become widespread – making new models estimable and becoming standard in some social science fields

How Bayesian Statistics Work

Bayesian methods rely on Bayes' Theorem

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)} \propto P(B \mid A)P(A)$$

Here:

- $P(A)$ is the prior distribution (pdf) of A (i.e., a Bayesian method)
- $P(B)$ is the marginal distribution (pdf) of B
- $P(B \mid A)$ is the conditional distribution (pdf) of B, given A
- $P(A \mid B)$ is the posterior distribution (pdf) of A, given B

A Live Bayesian Example

- Suppose we wanted to assess the probability of rolling a 1 on a 6-sided die:

$$p_1 = P(D = 1)$$

- We then collect a sample of data $\mathbf{X} = \{0, 1, 0, 1, 1\}$
 - These are independent tosses of the die
- The posterior distribution of the probability of a 1 conditional on the data is:

$$P(p_1 \mid \mathbf{X})$$

- We can determine this via Bayes theorem:

$$P(p_1 \mid \mathbf{X}) = \frac{P(\mathbf{X} \mid p_1)P(p_1)}{P(\mathbf{X})} \propto P(\mathbf{X} \mid p_1)P(p_1)$$

Defining the Likelihood Function

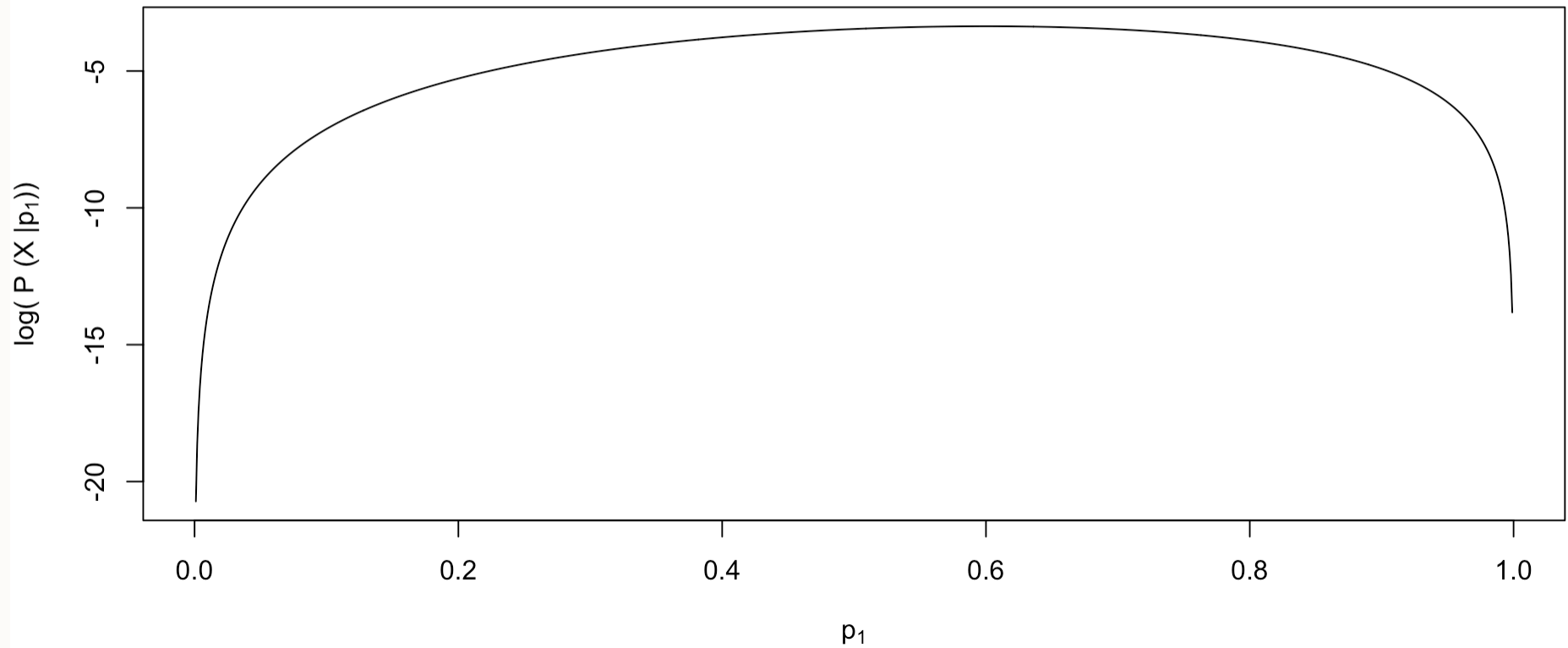
$$P(\mathbf{X} \mid p_1)$$

The likelihood of the data given the parameter:

$$P(\mathbf{X} \mid p_1) = \prod_{i=1}^N p_1^{X_i} (1 - p_1)^{(1-X_i)}$$

- Any given roll of the dice X_i is a Bernoulli variable $X_i \sim B(p_1)$
 - A “success” is defined by rolling a one
- The product in the likelihood function comes from each roll being independent
 - The outcome of a roll does not depend on previous or future rolls

Visualizing the Likelihood Function



Choosing the Prior Distribution for p_1

We must now pick the prior distribution of p_1 :

$$P(p_1)$$

- Our choice is subjective: Many distributions to choose from
- What we know is that for a “fair” die, the probability of rolling a one is $\frac{1}{6}$
 - But...probability is not a distribution
- Instead, let's consider a Beta distribution $p_1 \sim \text{Beta}(\alpha, \beta)$

The Beta Distribution

For parameters that range between zero and one (or two finite end points), the Beta distribution makes a good choice for a prior:

$$P(p_1) = \frac{(p_1)^{\alpha-1} (1 - p_1)^{\beta-1}}{B(\alpha, \beta)},$$

where:

$$B(\alpha, \beta) = \frac{\Gamma(\alpha) \Gamma(\beta)}{\Gamma(\alpha + \beta)},$$

and,

$$\Gamma(z) = \int_0^{\infty} t^{z-1} e^{-t} dt$$

More Beta Distribution

The Beta distribution has a mean of $\frac{\alpha}{\alpha+\beta}$

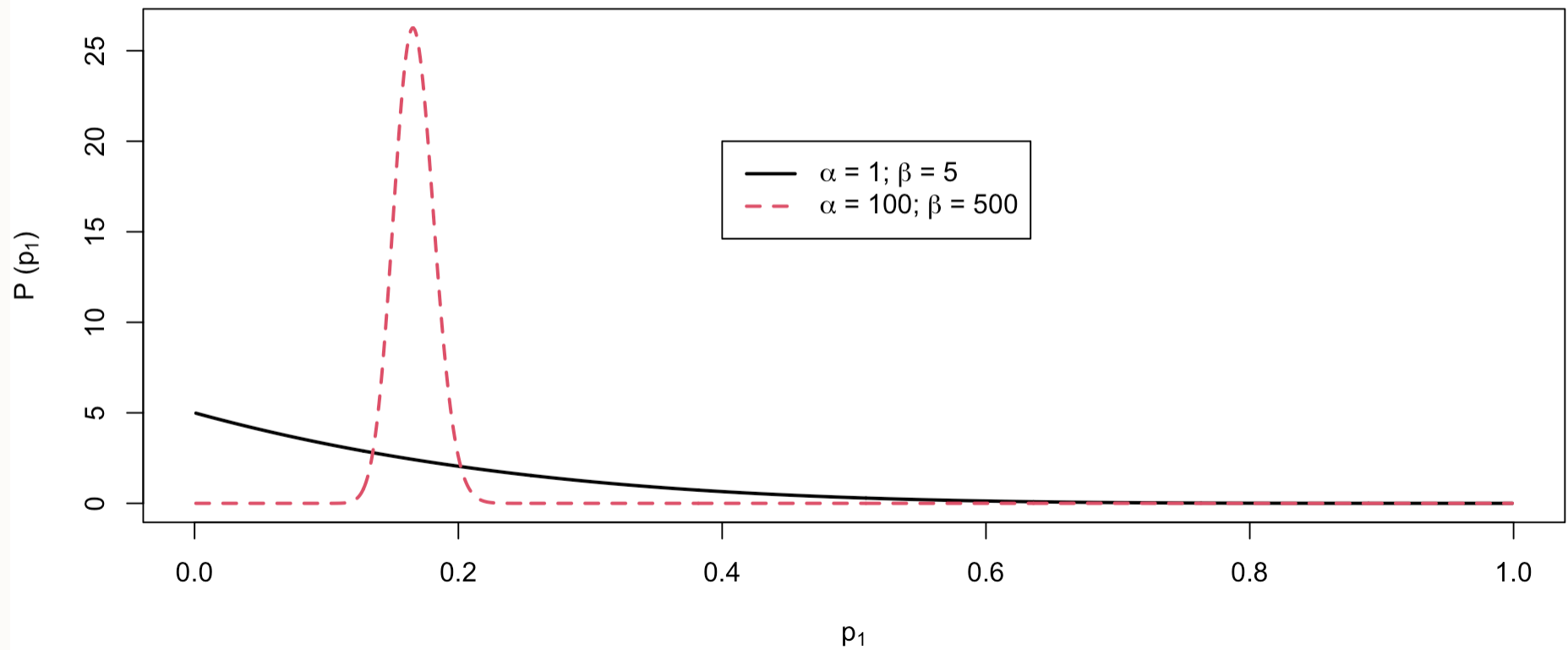
- The parameters α and β are called hyperparameters
 - Hyperparameters are parameters of prior distributions
- We can pick values of α and β to correspond to $\frac{1}{6}$
 - Many choices: $\alpha = 1$ and $\beta = 5$ have the same mean as $\alpha = 100$ and $\beta = 500$
- What is the difference?
 - How strongly we feel in our beliefs...as quantified by...

More More Beta Distribution

The Beta distribution has a variance of $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$

- Choosing $\alpha = 1$ and $\beta = 5$ yields a prior with mean $\frac{1}{6}$ and variance 0.02
- Choosing $\alpha = 100$ and $\beta = 500$ yields a prior with mean $\frac{1}{6}$ and variance 0.0002
- The smaller prior variance means the prior is more informative
 - Informative priors are those that have relatively small variances
 - Uninformative priors are those that have relatively large variances

Visualizing $P(p_1)$

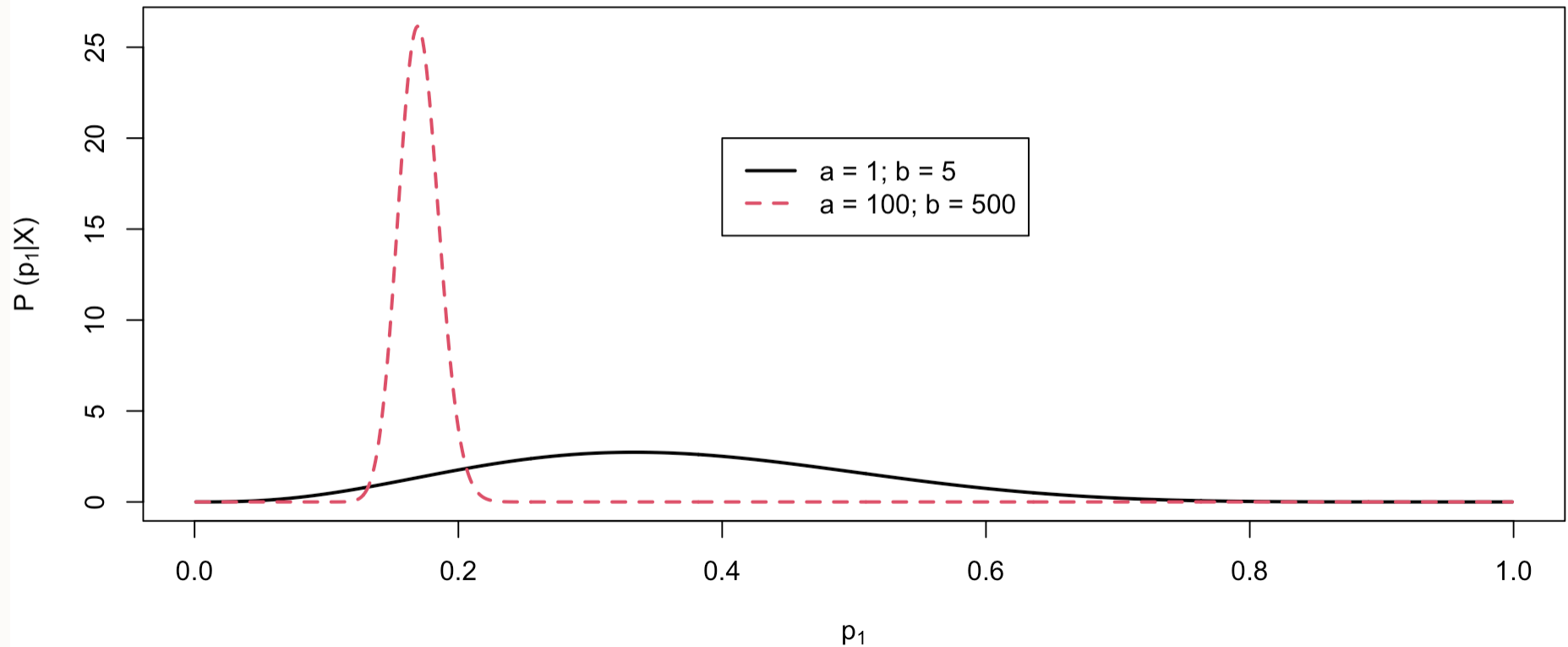


The Posterior Distribution

Choosing a Beta distribution for a prior for p_1 is *very* convenient

- When combined with Bernoulli (Binomial) data likelihood the posterior distribution can be derived analytically
- The posterior distribution is also a Beta distribution
 - $\alpha = a + \sum_{i=1}^N X_i$ (a is the hyperparameter of the prior distribution)
 - $\beta = b + N - \sum_{i=1}^N X_i$ (is the hyperparameter of the posterior distribution)
- The Beta prior is said to be a conjugate prior: A prior distribution that leads to a posterior distribution of the same family
 - Here, prior == Beta and posterior == Beta

Visualizing The Posterior Distribution



Bayesian Estimates are Summaries of the Posterior Distribution

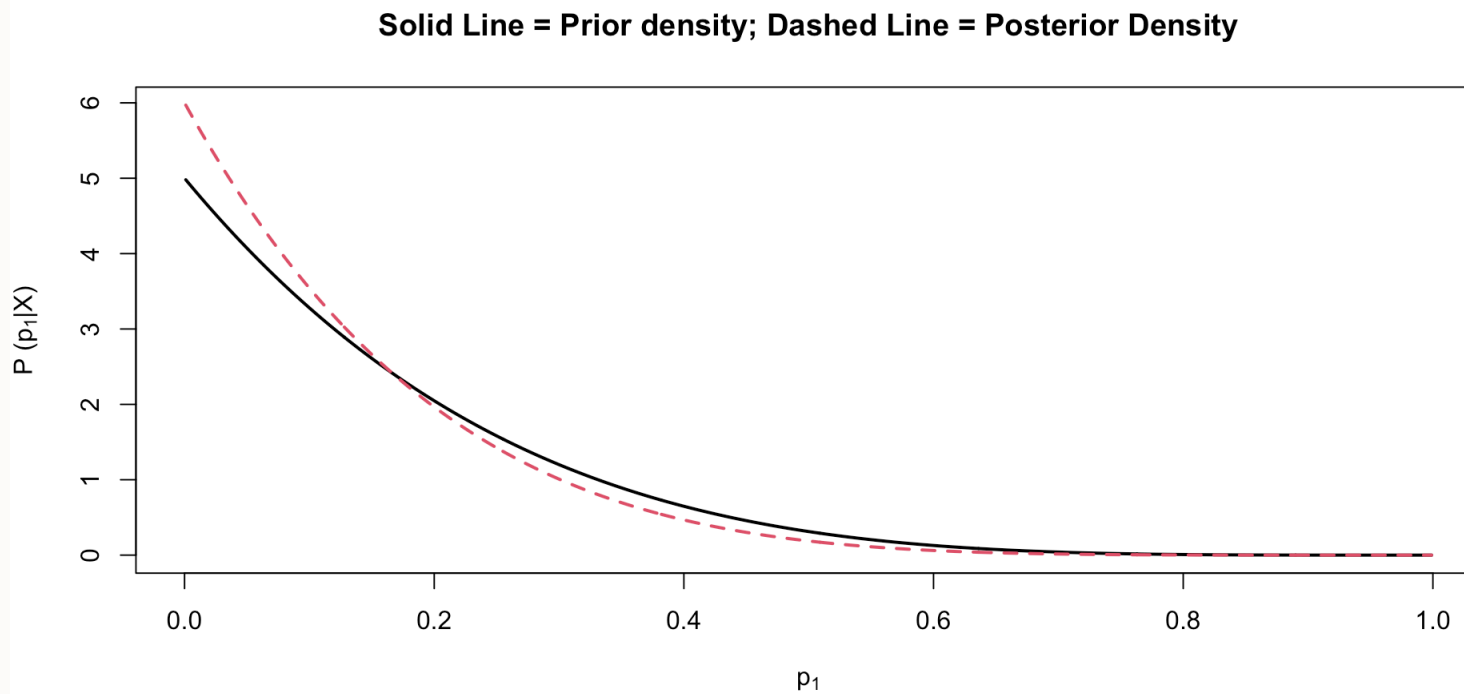
To determine the estimate of θ , we use summaries of the posterior distribution:

- With prior hyperparameters and $\mathbf{y}$
 - $\text{SD} = 0.1388659$
- With prior hyperparameters and $\mathbf{y}^*$
 - $\text{SD} = 0.0152679$
- The standard deviation (SD) of the posterior distribution is analogous to the standard error in frequentist statistics

Bayesian Updating

We can use the posterior distribution as a prior!

Let's roll a die to find out how...



Section Summary

This section was a very quick introduction to Bayesian concepts:

- prior distribution
 - hyperparameters
 - informative/uninformative
 - conjugate prior
- data likelihood
- posterior distribution
- Next we will discuss MCMC and Stan